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Title:Non-convex transfer subspace learning for unsupervised domain adaptation

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Abstract:Transfer subspace learning aims to learn robust subspace for the target domain by leveraging knowledge from the source domain. The traditional methods often adopt the convex norm to approximate the original sparse and low-rank constraints, which make the optimization problem be easily solved. However, such relax approximation leads to the performance deviation of the original non-convex model. In this paper, we propose a novel Non-convex Transfer Subspace Learning~(NTSL) method to provide a tighter approximation to the original sparse and low-rank constraints. Specifically, we design an objective function that leverages the Schatten p-norm and ℓ 2, p-norm to preserve the structure between the

NON-CONVEX TRANSFER SUBSPACE LEARNING FOR UNSUPERVISED DOMAIN ADAPTATION

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ABSTRACT

Transfer subspace learning aims to learn robust subspace for the target domain by leveraging knowledge from the source domain. The traditional methods often adopt the convex norm to approximate the original sparse and low-rank constraints, which make the optimization problem be easily solved. However, such relax approximation leads to the performance deviation of the original non-convex model. In this paper, we propose a novel Non-convex Transfer Subspace Learning (NTSL) method to provide a tighter approximation to the original sparse and low-rank constraints. Specifically, we design an objective function that leverages the Schatten p-norm and $\ell_{2,p}$-norm to preserve the structure between the source and target domains. With Schatten p-norm, the objective function better approximates the rank minimization problem than the nuclear norm and preserves the structure of domains. Besides, the $\ell_{2,p}$-norm can reduce the effect of noise and improve the robustness to outliers. Meanwhile, we develop an efficient algorithm to solve the non-convex minimization problem. Extensive experimental results on cross-domain tasks show the effectiveness of our proposed method.

***Index Terms*—** Domain adaptation, Subspace learning, Non-convex optimization, Schatten p-norm, Low-rank model

1. INTRODUCTION

The lack of labeled data and the price for labeling samples drive people to use labeled and relevant data from another distinct database or domain. The urgent needs boost the development of domain adaptation. Unsupervised domain adaptation aims at training classifiers with samples from source domain and then applies classifiers to unlabeled samples from target domain [1]. In unsupervised domain adaptation, the direct application of traditional statistical learning models probably obtains unsatisfactory performance due to the distribution shift problem, which is caused by the different distribution between domains. The transfer subspace learning approaches are proposed to tackle the distribution shift problem.

Recently, transfer subspace learning shows its potential in unsupervised domain adaptation. The basic approaches for transfer subspace learning seek subspace where the distribution divergence between domains gets decreased [2, 3] or the relationship among distributions are modeled [4, 5, 6, 7]. In the literature, there are two main ideas for transfer subspace learning. The first idea is based on manifold learning, which could effectively extract the global and local structure of data in domains. For instance, Geodesic Flow Kernel (GFK) [3] finds a geodesic from the source domain to the target domain based on the kernel methods. The other idea is based on the statistical feature alignment on the low dimensional subspace, which minimizes the divergence between two distributions. In this paper, we focus on the second type of transfer subspace learning methods.

Among the second type of methods, there are many transfer subspace learning methods, such as Transfer Subspace Learning (TSL) [2], Subspace Alignment (SA) [4], Low-rank Transfer Subspace Learning (LTSL) [5], and so on. TSL and SA, which are two representative methods of them, adopt subspace learning to obtain the statistical alignment of source domain and target domain. However, due to the lack of low-rank and sparse constraints, the alignment of source domain and target domain may be not robust under noise [5, 8]. Many studies [5, 6, 7, 9, 10] find that importing the low-rank and sparse constraints into transfer subspace learning could effectively improve the performance of domain adaption. Shao *et al.* [5] gave a concise proof of the boundary of the objective function for low-rank transfer subspace learning. Furthermore, Xu *et al.* [6] used the ℓ_1-norm sparse constraint to obtain the impressive improvement. The ideal low-rank and sparse constraints can be solved by the minimization of the rank and the $\ell_{2,0}$-norm. However, they are well known NP-hard problems and difficult to solve directly [11, 12]. Therefore, to solve the optimization problem efficiently, these traditional methods adopt nuclear norm and $\ell_{2,1}$-norm to replace them, respectively. And then they obtain the approximate optimal solution by solving the relaxed convex objective func-

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